SANDIP SINGH

Blockchain Developer | Escrow & Payments | Smart Contracts Engineer
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EXECUTIVE SUMMARY

Blockchain Developer with 5+ years of experience specialising in Ethereum-based payment gateway solutions, including escrow smart contracts, multi-currency settlement, and oracle-driven price conversions, with end-to-end front-end and back-end integration. Proven expertise in Solidity, gas optimisation, smart contract architecture, and unit/integration testing to deliver secure, scalable, and audit-ready deployments across Ethereum and Polygon networks. Experienced across multiple blockchain ecosystems, including Ethereum, Solana, Avalanche, and Polygon, with strong expertise in Solana development using Rust, covering Devnet setup, TPS benchmarking, and PoC execution.

EXPERIENCE

AUG 2024 – OCT 2025

BLOCKCHAIN DEVELOPER, AI TECH SOLUTIONS LIMITED.

- **Architected and implemented escrow smart contracts** to securely lock and release client and freelancer funds based on milestone-driven workflows, supporting **1,000+ on-chain escrow transactions** with deterministic and trustless settlement logic.
- **Designed, developed, and deployed Ethereum-based payment gateway modules** integrated with escrow contracts and oracle-driven pricing systems to enable secure **multi-currency transactions** and automated settlement workflows.
- **Led the design and implementation of gasless smart contract interactions** using meta-transaction patterns, significantly improving user experience by abstracting gas fees and enabling **non-custodial, frictionless payments**.
- **Collaborated closely with backend engineering teams** to implement decentralised payment workflows, integrating Solidity smart contracts with backend services using **Hardhat and Ethers.js**, supporting **hundreds of daily contract interactions**.
- **Developed comprehensive unit and integration test suites**, covering critical payment and escrow paths, achieving **90%+ test coverage** and ensuring smart contract **security, performance, and audit readiness**.
- **Optimised gas consumption and smart contract architecture**, reducing average transaction costs by **25–35%** through refactoring, storage optimisation, and efficient execution patterns, preparing contracts for **production deployment on Ethereum and Polygon test networks**.

JAN 2022 – JUNE 2024

BLOCKCHAIN DEVELOPER, GATHER SOFTWARE PVT. LTD.

- **Led the architecture and implementation of a forked Solana Layer-1 blockchain network**, delivering a custom native token with tailored **genesis configuration, consensus parameter tuning, and tokenomics design** aligned to network incentives and security.
- **Developed and deployed on-chain programs and smart contracts** on the custom Solana network, and **set up and managed a dedicated Devnet** to support protocol testing, validation, and iterative feature development.
- **Contributed to a distributed computing initiative**, designing a **blockchain-based marketplace for computational power**, enabling decentralised task execution, pricing, and settlement using on-chain coordination mechanisms.
- **Designed and deployed staking and masternode DApps** using **Solidity, Hardhat, and Web3.js**, improving platform usability, reward distribution logic, and overall user experience for network participants.
- **Integrated indexing and data-query layers using Subgraphs / Substreams**, enabling efficient on-chain data access for analytics, dashboards, and application-layer consumption.
- **Implemented modern Web3 front-end integrations** using **Wagmi and Viem**, improving wallet connectivity, contract interaction reliability, and developer experience across decentralised applications.

APR 2020 – OCT 2021

JUNIOR BLOCKCHAIN DEVELOPER, SODIO TECHNOLOGY PVT. LTD.

- **Conducted initial client requirement analysis and technical feasibility assessments**, translating business needs into blockchain solution designs and research-backed implementation plans.
- **Built front-end monitoring dashboards and web automation tooling** using **Node.js and Puppeteer** to track

network activity, validator behaviour, and system health, improving operational visibility and debugging efficiency.

- **Built and integrated RESTful backend services** to support smart contract interactions and enhance application usability, enabling seamless communication between the blockchain layer and user-facing interfaces.
- **Collaborated with cross-functional teams** to align smart contract functionality with backend APIs and frontend workflows, improving overall **user experience, performance, and system reliability**.

TECHNICAL SKILLS

- **Blockchain:** Smart contract development • DApp development • Smart Contract Security & Auditing • Gasless transactions • Oracles.
- **Programming:** Solidity • Rust • JavaScript/TypeScript.
- **Tools:** Hardhat • Anchor • Remix • Truffle • Foundry • Ethers.js • Web3.js • Next.js • Subgraphs/Substreams • Wagmi • Viem.
- **Backend/Automation:** Node.js • REST APIs.

PUBLICATION

Sandip Singh, M. Edison, "*A compact 3rd Order SIW Filter with Improved Upper Band Performance for Ku-Band Application*", International Conference on Innovation in Electronics, Signal Processing and Communication (IESC), IEEE 2019.

EDUCATION

JULY 2016 - JUNE 2018

MASTER OF TECHNOLOGY, NERIST, ARUNACHAL PRADESH.

7.74 GPA in Electronics and Communication.

JULY 2012 - JUNE 2016

BACHELOR OF TECHNOLOGY, SHIATS, PRAYAGRAJ, UTTAR PRADESH.

7.3 GPA in Electronics and Communication.